ISBA 2411 · The Retrieval Hackathon — Facilitator Run-Sheet

~60 min

Meaning vs. Words. Teams build a search engine over a shared 800-passage corpus, compete on a live leaderboard, then a paraphrase "curveball" round flips the board. Board line: *"Keyword search matches words. Semantic search matches meaning. Which wins depends on the query."*

⌖ PRE-FLIGHT (before class)

- Pre-run the starter notebook once so the dataset + models are cached; note the baseline (main MRR 0.78, curveball 0.47).
- Open the leaderboard sheet and set sharing to **Anyone with the link · Editor**; confirm the link is in the notebook's last cell.
- Tell teams to use a **GPU runtime** if they try bigger embedding models (Runtime > Change runtime type > GPU).
- Kit lives in the repo: `Week 4/search_hackathon/` (notebook, data, rubric).

0:00–0:05 1 · Kickoff

DO	Hand out the notebook (Open in Colab badge). Everyone runs Section 0 + the TF-IDF baseline cell — that's the number to beat: **main MRR ~0.78**.
SAY	"Same corpus, same queries, same metric for everyone — how you build the retriever is up to you. Beat the keyword baseline, then beat each other. The board is live."

0:05–0:35 2 · Build sprint

DO	Teams build and submit to the live leaderboard as they iterate. Circulate. Nudge them through the ladder: TF-IDF → BM25 → dense embeddings → hybrid → re-rank.
WATCH	Early on, **keyword/BM25 teams will lead** the main-round column — that's expected and correct. Don't spoil why; let the curveball reveal it.
SAY	"Keyword search is winning the main round. Hold that thought — it's a strong baseline, and that surprises people. In a minute we test something it can't do."

0:35–0:45 3 · The curveball reveal

DO	Point the room at the **Curveball MRR** column (already computed in every submission). Announce it now counts. These are paraphrased queries — same meaning, different words.
WATCH	The board reorders: dense/semantic teams surge (curveball MRR ~0.66) while keyword-only teams sink (~0.47). This is the lesson made physical.
SAY	"The words changed but the meaning didn't. Keyword search needs the words; semantic search follows the meaning. That gap is the whole reason embeddings exist."

0:45–0:56 4 · Demos (~2 min / team)

DO	Each team live-demos **one query their engine nails that the keyword baseline misses**, and states its quality-vs-cost trade-off.
SAY	"Don't just tell me your score — show me a search the baseline gets wrong and yours gets right, and tell me if you'd ship it."

0:56–1:00 5 · Awards + read the frontier

DO

Hand out several awards (below), then plot the board: quality vs. build-time. LogReg-of-search (TF-IDF) is cheap; hybrid usually tops the frontier.

Awards: Best Recall · Fastest Index · Best Value (frontier) · Biggest Comeback (main→curveball) · Meaning over Words · Best Pitch / People's Choice.

SAY

"Two rounds, one lesson: neither method wins everywhere. The best system usually fuses them — and the person who can tell which to trust for a given job is the one who ships good search."

IF IT BREAKS

Model download or GPU stalls a team → they fall back to TF-IDF + BM25 (both CPU, no download) and still compete on the main round. The dataset loads from the repo, so nothing depends on a live Hugging Face pull.

Validated numbers (this dataset): TF-IDF main MRR 0.782 / curveball 0.474 · Dense MiniLM main 0.737 / curveball 0.664 · hybrid expected to top both. Keyword wins literal queries; semantic wins paraphrases; nothing is saturated, so there is real room to climb.